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Mitigating Wellbore Stability Risks in Hydraulic Fracturing: A Case Study from the Structurally Complex Unconventional Tight Middle Marrat Formation

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Abstract

This study presents key lessons learned from hydraulic fracturing challenges in the unconventional tight carbonate Middle Marrat Formation in North Kuwait Jurassic Gas fields. The horizontal appraisal well was drilled through two porous geo-bodies separated by a tight layer, intersecting 17 faults, which introduced structural complexity and wellbore stability risks. A multi-stage hydraulic fracturing approach was applied, integrating 3D Deep Fracture Mapping (3DDFM), mechanical earth modeling (MEM), and real-time formation imaging to optimize stimulation. This paper will demonstrate how integration of reservoir characterization diagnostics improves fracture connectivity mapping, stimulation design and diversion efficiency, understand structural challenges and how all such factors affect well productivity in addition to plug and perf operational challenges in horizontal monobore wells.

To overcome uncertainties in reservoir connectivity, a slim Dipole Sonic (SDST) tool was deployed in the 4.5 inch cased lateral, conveyed by a slim tractor for acoustic velocity measurements. 3DDFM provided insights into far-field fracture connectivity between the two porous geo-bodies, aiding in stimulation design. A four-stage acid fracturing treatment was executed using Single Phase Retarded Acid System (SPRAS) and viscoelastic acid diverter, with real-time multi-finger caliper (MFC) logging due to restrictions in setting plugs. The number of fracturing stages was reduced after detecting liner deformations at locations of suspected active faults.

The 3DDFM confirmed the structural dip of the geo-bodies and identified sub-layer connectivity through faulted fractures. Reservoir geosteering inversion further validated reservoir heterogeneity, but stimulation planning required additional stress profiling. MEM analysis confirmed high stress anisotropy, with closure pressures changes up to 2,700 psi.

During execution, MFC revealed severe liner deformations at active fault locations, correlating directly with faults mapped by 3DDFM. These restrictions prevented full coiled tubing access during post-fracturing. Datafrac results showed low fluid efficiency, suggesting high fluid leak-off into natural fractures. However,

particulate diverters successfully controlled fracture height growth, with pressure responses confirming effective diversion.

This case study highlights the importance of integrating geophysical and geomechanical data in complex carbonate formations, demonstrating how fault mapping, mechanical modeling, and real-time diagnostics can mitigate risks and improve hydraulic fracturing success.

This study represents the first integration of 3DDFM, Reservoir geosteering inversion, and MEM analysis for stimulation optimization in the unconventional Upper Middle Marrat Formation. The findings provide critical insights into fault-controlled fracture propagation, wellbore stability risks, and acid fracturing efficiency in tight carbonates. The lessons learned establish best practices for planning multi-stage hydraulic fracturing in structurally complex reservoirs, ensuring better connectivity, reduced fluid loss, and improved long-term well performance.

Introduction to North Kuwait Jurassic Reservoirs

The Kuwait North Gas Field asset includes carbonate reservoirs with near critical gas and volatile oil fluids. These reservoirs are complex in terms of reservoir quality and fluid and rock properties and are very challenging with regards to production both upstream and downstream with typical depths from 14,000 to 18,000 ft MD (14,000 – 15,000 ft TVD). The produced fluid is sour and contains high H_2S (up to15%) and CO_2 (up to 6%). The reservoir pressure ranges from virgin high pressure to depleted (down to <40% of its original level) with an average reservoir temperature of 275°F.

The recovery from these reservoirs is uncertain if a conventional development strategy is applied. Therefore, a dedicated full development plan applying integrated upstream and downstream technologies is critically important.

The main pay zone (Middle Marrat reservoir) is a tight limestone (Calcite > 90%) with very low porosity and permeability, streaked with layers of dolomite that have superior rock properties. A pressure transient analysis (PTA) that was performed in various wells showed dual porosity behavior with low matrix permeability as storage volume. The average matrix permeability is less than 1 mD (Al-Saeed et al 2017).

Figure 1 and Figure 2 show the Kuwait Jurassic Stratigraphy and typical log showing high permeability contrast among reservoir layers, respectively.

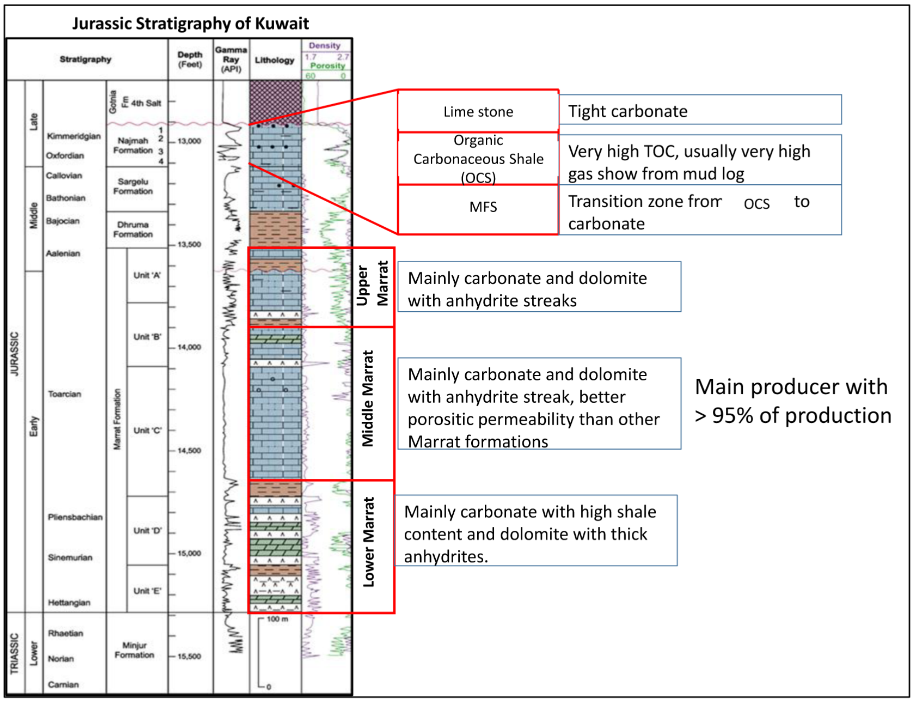

Figure 1—Kuwait Jurassic Stratigraphy

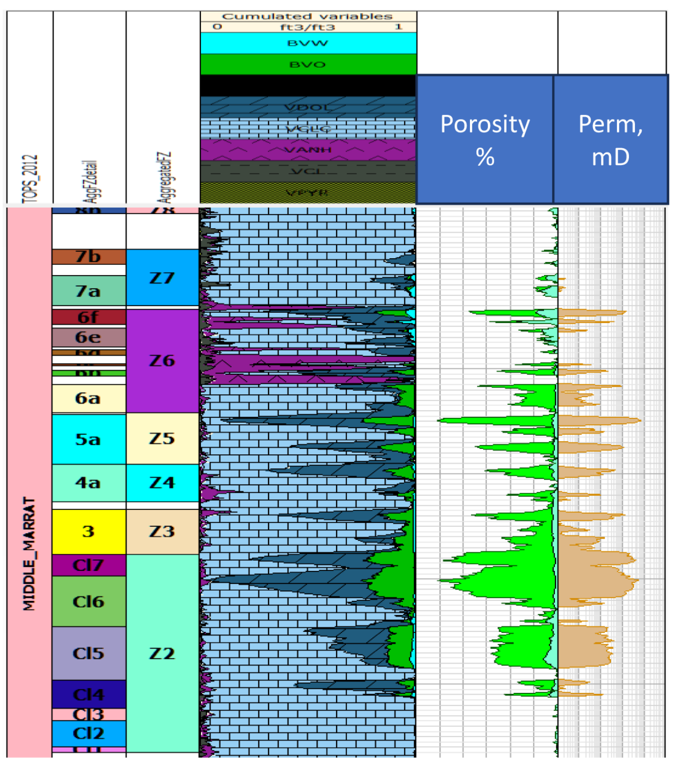

Figure 2—Typical log showing high permeability contrast among reservoir layers

North Kuwait Jurassic Reservoirs Description/Classification

1. **Middle Marrat**

   "The Middle Marrat reservoirs consist of dolomitic reservoir units with gross thickness ranging from 10-50 ft. Some limestone stringers are likely. Typically, these reservoirs are interbedded with tight limestone/anhydrite layers with variable thickness from few to up-to 60 ft. Reservoir properties obtained from e-log and core analysis show low reservoir qualities with matrix effective porosity from 5-10%, and very low permeability, typically < 0.01 mD.

The top section of the Middle Marrat formation is sub-divided in to Flow Zones 7 and 8 (FZ-7 &8) that are considered as extremely tight High-Pressure High-Temperature unconventional layers specially in the platform area of Sabriah (SA) field, where the subject case study of this paper is located.

2. **Lower Marrat**

   Five reservoir layers have been identified in the Lower Marrat: Dolomite 1-5. These units mainly comprise porous dolomite, but locally also contain tight limestone and varying amounts of clay, which becomes prominent towards the top of the formation (top of the zone-5), close to the boundary with the Middle Marrat. These units vary significantly in thickness and reservoir quality, with Dol-3 and Dol-5 being the thickest and Dol-1 the most porous. Thickness within the single units is quite constant; porosity and N/G are variable but show some degree of interdependence.

3. **Upper Marrat**

   Reservoir quality in the Upper Marrat Formation is generally tight with low natural fractures. Based on log observations, four reservoir units have been identified (Dol-1 to Dol-4) in the lower part of the Upper Marrat Formation. They comprise varying amounts of porous dolomite but also contain high amounts of interbedded anhydrite, limestone and clays. All four reservoir units are relatively thin (4-10ft) and of similar reservoir quality (av. porosity 6-8%). The most attractive intervals are Dol-2, 3 and 4, with Dol-2 and Dol-3 having the highest net thickness and Dol-4 having a higher N/G and porosity than Dol-1 The thickness within the single reservoir units is quite uniform, porosity and N/G show some degree of interdependence.

4. **Najmah Sargelu**

   The **Najmah** Formation of North Kuwait is a hydrocarbon-bearing reservoir in all 6 North Kuwait fields. It is made up of tight carbonates, with porosity and permeability provided mainly by fractures, and matrix porosity limited to thin intervals.

   The **<u>Najmah Limestone</u>** is a tight, fractured limestone reservoir, with an average thickness of 44ft and with matrix porosity in thin layers, typically less than 5ft thick. Natural fractures typically increase in density at few feet from the top and from base of limestone, closer to contact with different strength of rock in neighboring formations. Petrophysical re-evaluation confirmed earlier porosity estimates averaging 4.5%; average permeability (0.01mD); and average hydrocarbon saturation (53 %). Production is characterized by good initial well rates followed by rapid decline (within months) and early water production in some cases. Productivity appears largely associated with the presence of open fractures.

   The **<u>Najmah Organic Carbonaceous Shale (OCA)</u>** member is black calcareous shale (marl) with a high organic content (av. 7-12%). This layer also extends regionally across Kuwait and drowns into surrounding areas, roughly at time equivalent with the Hanifa Formation, which was deposited across Middle East (e.g., Saudi Arabia, Qatar) in the Late Jurassic.

   The **<u>MFS</u>** forms the gradual transition from the kerogen rich Najmah Kerogen Formation to the Sargelu very tight limestones. The MFS zone consists of carbonate layers (skeletal limestones with minor dolomites) interbedded with black Kerogen/organic rich layers (TOC 2-8%). Carbonate layers thickness increases downwards, while kerogen layers are thinning and completely absent at the base of MFS. The kerogen layers are very similar to overlaying Najmah Kerogen zone, with almost no fractures, while interbedded skeletal limestones are often heavily fractured, particularly in the slightly grainier parts of the rock, which may support production. The limestone and kerogen are tight, with very low matrix porosity < 4%, and permeability from 0 – 0.006mD".

Tight Middle Marrat (FZ-7&8) Development Plan

The SA platform lies beyond the carbonate platform edge and hence comprises low porosity and permeability limestones with negligible development of dolomite. To develop this wide area but low Hydrocarbon density resources require long horizontal wells with multistage hydraulic fracturing stimulation treatments. The long horizontal are needed to maximize reservoir contact and access to enough volume and the multistage fracturing to achieve sufficient KH to enable achieving commercial flow rates from such extremely tight reservoirs. The potential of FZ-7&8 has been tested in couple of vertical wells in SA platform area however it did not show sustained commercial production. Therefore, and due to the relatively high associated reserves, the development plan has been set to drill long horizontal wells with multistage hydraulic fracturing stimulation. The subject study well is the appraisal horizontal well which has been drilled in FZ-7 to evaluate the reservoir production potential in SA platform area.

Introduction to Cemented Casing Deformation

Unconventional resources, characterized by their exceptionally low permeability, demand technically novel, cost-effective, and scalable solutions for their development in an economically viable manner. Horizontal drilling combined with Multi-Stage Hydraulic Fracturing (MSHF) has emerged as the key approach to unlocking this resource potential economically. However, amid this transformative progress, an inconvenient complication that could potentially hinder the economic viability has begun to emerge and be experienced. Specifically, there are an increasing number of reports of casing deformation taking place, predominantly located within the lateral section of these wells, which can occur prior to, during, or after the execution of these MSHF operations. These deformations, although not comprising a well integrity event or indeed posing any form of direct threat to safety and/or the environment, do create potential obstructions within the casing, impeding subsequent planned hydraulic fracturing stages and potentially limiting access to the well sections deeper than these restrictions. The result of this can be sub-optimal stimulation coverage and distribution, a mounting level of Non-Productive Time (NPT), increasing cumulative costs and reduced Estimated Ultimate Recovery (EUR). A combination of these challenges can collectively erode the overall economic delivery and from this perspective addressing casing deformation effectively becomes a crucial and imperative factor for enhancing overall economic performance of the unconventional resource sector.

To gain comprehensive insights into the issue of casing deformation during the execution of MSHF in unconventional reservoirs, an extensive literature review was conducted (Ref. SPE-217822-MS, J. A. Uribe-Patino et. Al, HFTC, 2024).

Symptoms and Manifestations of Casing Deformations

Casing deformation occurrences exhibit a broad spectrum of individual characteristics, manner of manifestation and ultimately their underlying root causes. While this complexity may lead to a diverse range of observations across multiple wells and studies, these insights serve as valuable guides for comprehending the intricacies of casing deformation behaviour. It falls upon the researchers to meticulously investigate each of the cases that are cited within this document and thereby assess the specific attributes, in each case, that directly contribute to the problem.

Four types of potential root-causes for casing deformation have been previously suggested and defined (Casero & Rylance, 2020), which are Type I (casing and tools), Type II (cementing quality), Type III (localised tectonics and formation rock properties), and Type IV (operations), as shown in Figure 3. (Ref. SPE-217822-MS, J. A. Uribe-Patino et. al, HFTC, 2024).

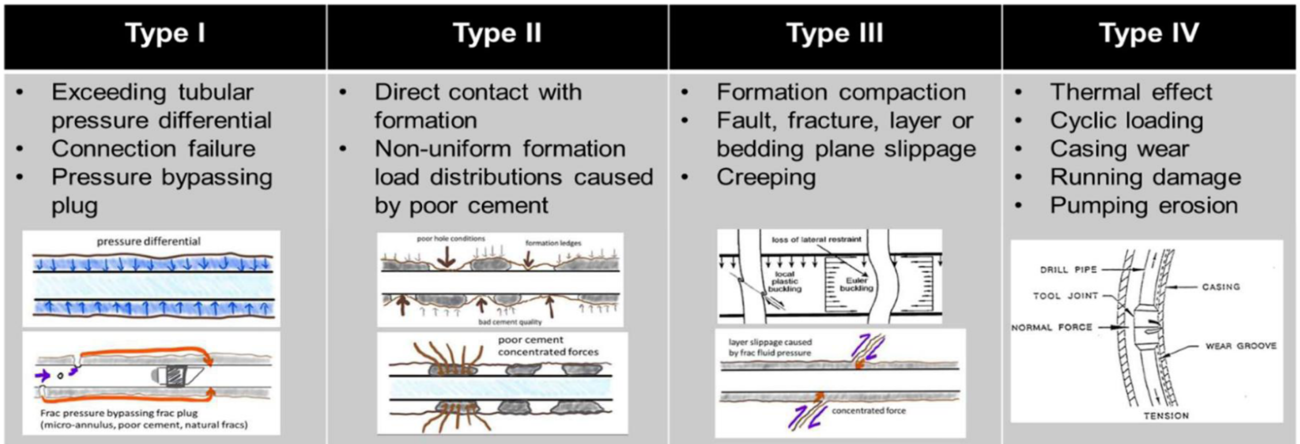

Figure 3—Four types of root-causes for casing deformation have been defined, which are Type I (casing and tools), Type II (cementing quality), Type III (tectonics and rock properties), and Type IV (operations). Adapted from Casero and Rylance (2020).

Among the four root-cause categories depicted in Figure 3, Type III, which relates to tectonics and rock properties, emerges as the prevailing category. In fact, it constitutes a substantial 60%, as indicated in Figure 4. This statistic firmly underscores the notable significance of Type III root-causes in the context of casing deformation during MSHF.

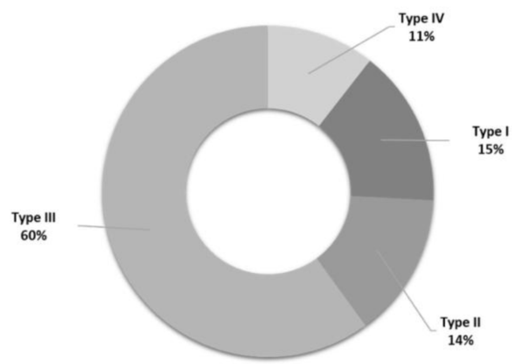

Figure 4—Summary of the analysis for root-causes of casing deformation (Ref. SPE-217822-MS, J. A. Uribe-Patino et. al, HFTC, 2024).

Horizontal Well SA-XX in Tight Middle Marrat SA Field Platform (Case Study)

Introduction to Well SA-XX

Well SA-XX is a Middle Marrat appraisal well drilled in the central part of the SA platform. It was drilled as a horizontal well (89.5 deg @ TD) with FZ #7 (unit b) in the Middle Marrat as the primary target. Final TD of this well was 19,035'MD. The total lateral length is 2656'. The wellbore intersected a number of small, sub-seismic scale faults at depths. The maximum angle observed in the well is 91.55 deg @ 18.316' MD and the maximum DLS is 6.4 deg/100' @ 16,140' MD. The initial plan for this well was to drill as 10,000 ft lateral in the tight layer separating FZ-7 and FZ-8. However, due to drilling challenges of the first section ~500 ft, the first lateral has been isolated and sidetrack decided to place the lateral in FZ-7b which deemed better dolomitic quality from offset vertical and highly deviated wells. Therefore, the well has been landed in FZ-7b with total lateral length of 2656 ft. The well trajectory did not cut any seismic fault. The main objective from relatively long lateral that: lateral increases chance of hitting discontinuous pay zones, model shows hydraulic fractures can penetrate multiple flow zones, greater access to the reservoir ensures higher recovery factors.

- **Well Location** (Figure 5):
  - Well located in enough space for an extend reach horizontal well
  - Away from offset wells and seismic faults
  - Structurally benign area, few expected faults/fractures
  - Well azimuth as close to field Sh min as possible while avoiding offset wells
  - Hydraulic fractures should not impact offset wells

- **Well Completion:**
  The well was planned initially to be completed with 4.5″ cemented sleeves multistage fracturing completion, however, due to unavailability of the cemented sleeves completion, the well has been completed as monobore with 4.5″ tubing and 4.5″ cemented liner in the reservoir section to enable rig-less plug and perf technique for multistage fracturing stimulation (Figure 6).

- **Multistage Fracturing: Staging Design, Planning, and Execution**
  The stimulation design has went through comprehensive analysis for the fracturing stimulation type (propped vs. acid fracturing). The final stimulation design considered acid fracturing due to the following analysis summary:
  - The lateral will be placed in higher stress layer (FZ7) compared to total zone of interest (FZ7/8).
  - Proposed fracs to cover this section and extend to FZ7 & FZ8 above and below.
  - This could cause a high risk in proppant screen-out due to pinch points around the wellbore cause by the high stress.
  - NWB tortuosity will most likely be high; therefore, it more difficult to sustain a high rate for sufficient fracture width to avoid screen-out in case of propped fracturing.
  - Therefore, the risk of screen-out is eliminated in acid fracturing case.
  - The risk of pinch-out still exists in acid fracturing case, however, with less impact since the acid will react and etch the formation (retarded acid highly recommended in this case).

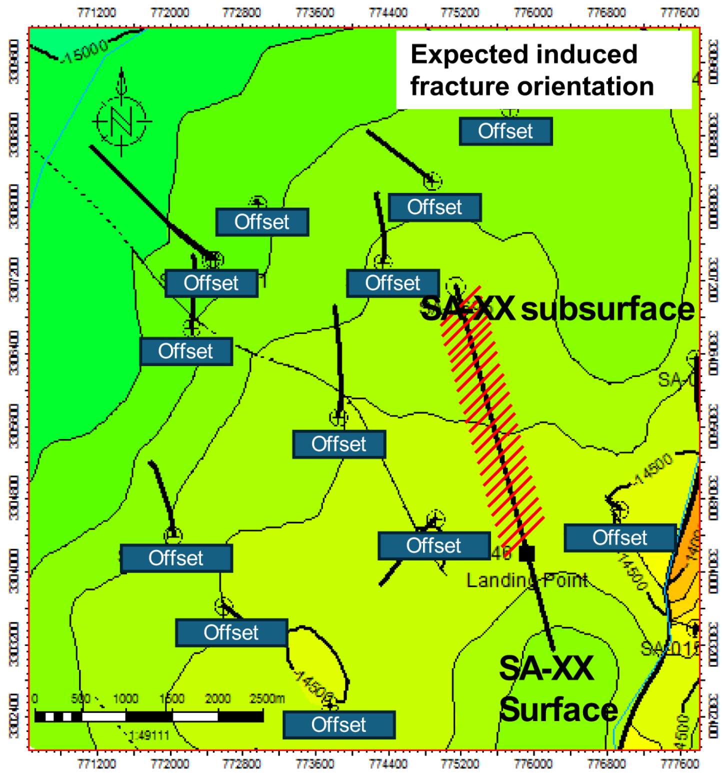

Figure 5—SA-XX Well Location and offsets

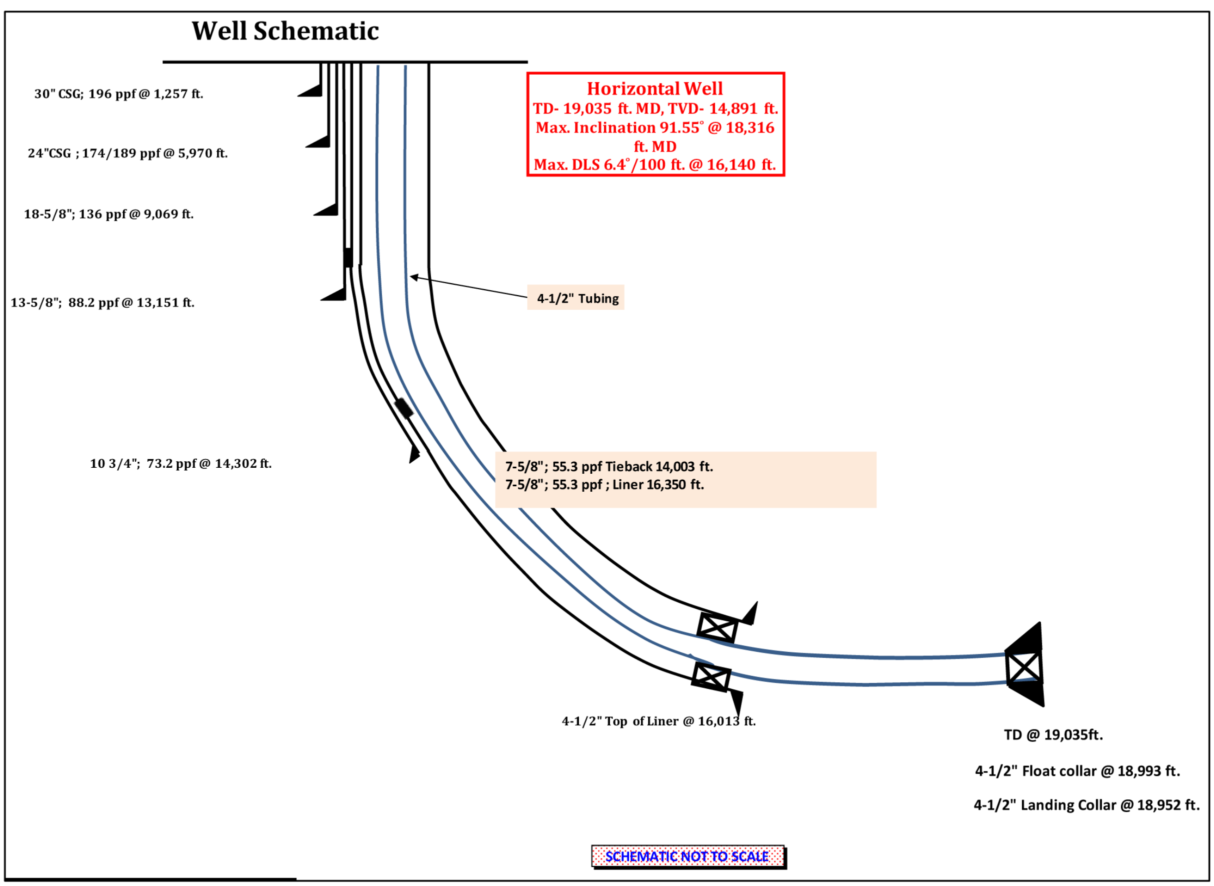

Figure 6—SA-XX Well Schematic post completion and prior plug and perf

Stimulation Staging Design, Execution and Outcome. A multi-stage acid fracturing stimulation was designed and executed in Well SA-XX, targeting the deep Middle Marrat Jurassic carbonate reservoir known for its tight matrix and high closure stress, which demand aggressive stimulation to achieve commercial production in Kuwait's Sabriya Field (TVD ≈ 14,900 ft). The treatment incorporated advanced fluid systems – a 15% single-phase retarded acid, a 15% viscoelastic diverting acid, and a biodegradable large size particulate diverter– to maximize reservoir contact across three stages from ~17,540 ft to 18,785 ft MD. Each stage consisted of repeated pad–acid–diverter cycles to sequentially stimulate multiple perforation clusters, with pump rates up to 37 bpm and treating pressures nearing 13,000 psi.

Data diagnostics (mini-frac/step-rate tests) were conducted prior to each main treatment, yielding closure gradients of 0.81–0.94 psi/ft and informing treatment parameters. The staged treatments were pumped as designed, achieving effective acid fracture propagation and diversion. Notably, measured pressure responses of +150–200 psi during biodegradable diverter pill deployment in Stages 2 and 3 confirmed successful near-wellbore diversion, whereas a weaker response in Stage 1 led to increasing diverter concentration in subsequent stages. Instantaneous shut-in pressures (ISIP) of ~9,300–9,600 psi (surface) were recorded, corresponding to bottomhole gradients ~1.05–1.08 psi/ft with fluid efficiency of ~40% from mini-frac.

Well SA-XX in Sabriya Field was drilled to intercept multiple productive layers within the Middle Marrat, completed with three sets of perforations separated vertically by several hundred feet. Given true vertical depths approaching 15,000 ft and expected high bottomhole static pressures, an acid fracturing treatment approach was selected to stimulate the low-permeability limestone creating conductive etched channels along fracture faces, but ensuring deep acid penetration and uniform coverage of multiple zones is challenging in thick carbonates.

Stimulation Strategy

To maximize reservoir contact, a multi-stage treatment was planned. Each stage would target one cluster of perforations, isolated from the others, and employ chemical diversion to sequentially stimulate the sub-intervals within that stage. A single-phase retarded acid (SPRA) system was chosen as the primary fluid to achieve deeper live-acid penetration before spending. Unlike conventional emulsified acids, the single-phase retarded acid contains a fully water-soluble retarding agent that slows the reaction rate without forming a separate hydrocarbon phase. This yields deeper wormholing and fracture etching, as well as lower friction pressure that enables higher pump rates with less resources requirement during fracturing. The retarded acid is expected to propagate fractures farther into the formation before spending, thereby extending the effective stimulated reservoir volume. To improve stimulation coverage across heterogeneous intervals, a viscoelastic diverting acid was incorporated. VDA is a self-diverting acid system (15% HCl with viscoelastic surfactant) that remains low viscosity while spending, then rapidly increases in viscosity in situ upon reaction with carbonate. This behavior allows VDA to temporarily plug high-permeability streaks or dominant fractures as acid spending occurs, diverting subsequent acid to lesser-treated zones. The VDA is polymer-free and degrades naturally upon contact with reservoir fluids, avoiding residual damage in the formation. In acid fracturing, the VDA can also aid leakoff control and improve fluid efficiency by reducing acid loss into the formation once it gels. As an additional near-wellbore diversion measure, a biodegradable large size particulate diverter (DLPD) pill was deployed during the treatments. This pill is part of an engineered diversion service that uses a blend of degradable fibers and multimodal solid particles to bridge fractures of various widths and create an impermeable, temporary plug. The fibers help suspend and distribute the particles, enabling the mixture to seal off the fracture or perforation tunnels effectively. Once in place, these fiber-enhanced pills impose a back-pressure, diverting subsequent fluid into untreated zones; they then dissolve over time (or with exposure to reservoir fluids) leaving no permanent blockage.

The large size particulate diverter was selected to augment the diversion provided by VDA, especially for near-wellbore faults and fractures near the fracture initiation points and between acid cycles. This chemical diversion approach was preferred over mechanical isolation within each stage to simplify operations and avoid costly multiple packer deployments and workovers. Mechanical isolation was only used between stages (via bridge plugs) to ensure each stage was treated independently.

Well and Reservoir Considerations

The treatments were designed with recognition of the high closure stress and potential variation in stress between sub-layers of the Middle Marrat. Prior well tests indicated a fracture gradient on the order of ~0.9–1.0 psi/ft in this reservoir. The depth range of Stage 1 (around 18,700 ft MD, ~14,800 ft TVD) suggested it would experience the highest stresses, while Stage 3 (17,550 ft MD, ~14,000 ft TVD) might have slightly lower overburden stress. Indeed, as shown later in step-rate tests, the upper Stage 3 interval exhibited a lower closure gradient (~0.81 psi/ft) compared to ~0.94 psi/ft in Stage 2. These differences implied that fracturing pressure and fluid leakoff characteristics could vary by stage. Furthermore, carbonate formations are known to contain natural fractures and vuggy porosity which can cause significant acid leakoff. Thus, pad fluid volume and pump rates were carefully optimized using preliminary DataFRAC tests to achieve an effective balance between creating a sizable fracture and avoiding rapid acid spend due to excessive leakoff.

Methodology and Treatment Design

Pre-Stimulation Diagnostics and Design Calibration

Each stage was preceded by diagnostic injections to characterize the formation's fracturing behavior and to calibrate the treatment design. For Stage 1, a DataFRAC sequence was performed, including a small-volume acid injection (breakdown) and a step-rate/fracture calibration test. These injections provided initial estimates of closure pressure, fracture gradient, and fluid leakoff characteristics. The breakdown test in Stage 1 indicated a closure pressure of ~12,465 psi (0.84 psi/ft gradient) for the lower Middle Marrat zone. The net pressure decline analysis and G-function plot from this mini-frac indicated a fluid efficiency of ~40%, i.e. about 40% of the pad fluid remained in the created fracture before closure. This relatively moderate fluid efficiency (for a carbonate acid frac) suggested significant leakoff, not unexpected given the formation's permeability and natural fractures. Consequently, the main treatment pad volume for Stage 1 was adjusted upward to account for this leakoff, ensuring sufficient fluid to open a fracture of desired length. Additionally, step-rate testing determined the fracture "extension pressure" and the rate above which the fracture would propagate. In Stage 1, fracture extension occurred at an estimated bottomhole pressure of ~16,060 psi at only ~2 bpm (due to the small injection), confirming that fracturing would initiate at high pressure but could be maintained at moderate rates once open. Stages 2 and 3, underwent abbreviated diagnostics. Before the main acid treatments, injectivity tests and step-rate tests (SRT) were conducted for these stages using slickwater (2% KCl brine) to minimize the overall fluid introduced to the reservoir to mitigate additional damage. For Stage 2, rates from 3 bpm up to 15 bpm were trialed. The SRT identified a clear change in pressure behavior at ~6 bpm, corresponding to a bottomhole extension pressure ~15,839 psi, and an estimated closure gradient of ~0.94 psi/ft (14,003 psi closure, at 14,888 ft TVD). Stage 3's SRT (rates 1–15 bpm) showed a lower fracture extension pressure ~12,319 psi and closure ~12,035 psi (0.81 psi/ft at ~14,000 ft TVD). The frictional pressure drops through the tubing/casing were also measured during these tests – for example, pumping 15 bpm of slickwater yielded ~2,300–2,500 psi friction. This led to the adjustment of maximum allowable surface pressure (MASP) limits during the main jobs, given the well's completion and pressure rating.

Fluid Systems and Pumping Schedule

The primary fluids used in the acid fracturing treatment by function were:

Pad Fluid

a customizable borate crosslinked delayed polymer-based fluid used to initiate and propagate the fracture. This fluid was pumped ahead of acid in each cycle to create fracture width for acid to penetrate. A total of ~6,180 bbl of pad was used across Stages 1–3. The pad fluid was formulated with a temperature-stable crosslinker formula (effective >200 °F) and included a friction reducer to allow high-rate pumping.

Main Acid

15% Single Phase Retarded Acid. In each cycle, after the pad established the fracture was pumped to etch the fracture face remaining as a single aqueous phase and contains a proprietary retarder that significantly slows the reaction rate with carbonate, roughly cutting the acid spending rate to allow deeper penetration compared to neat HCl. This extends fracture etching farther into the formation and improves acid utilization. It also yields a lower viscosity, enabling lower friction pressure and thus higher injection rates in long tubulars. Across the three stages, ~5,710 bbl of single-phase retarded acid was pumped.

Diverting Acid

VDA-15 (15% HCl with viscoelastic surfactant polymer-free diverter). Following each main acid slug, a volume of VDA was injected. In situ, the VDA acid begins to gel once a portion of the acid is spent, rapidly increasing in viscosity and thereby diverting subsequent fluid by preferentially flowing into lesser-treated zones. The VDA's viscosity buildup also aids in controlling fluid leakoff into the freshly acid-etched fractures, effectively improving the fluid efficiency of the treatment. A total of ~2,530 bbl of VDA was employed as the chemical diverter over all stages. Notably, VDA is self-breaking – the gelled acid will break, and flow back once exposed to reservoir fluids, leaving minimal residue.

Large Size Biodegradable Particulate Diverter

After a set number of acid/VDA cycles, a concentrated diverter pill was pumped. The pill comprised a large size slurry of degradable fibers and particles (in a light carrier fluid) designed to bridge at the fracture entry or within the near-wellbore region. The fibers (which withstand bottomhole temperatures above 200 °F) and acid-soluble particles form a matrix that temporarily plugs the fracture when deposited. This plug is meant to isolate the recently acidized zone, forcing subsequent pad and acid to initiate a new fracture (or open a different set of perforations) in the next interval. With time and based on design objectives the diverter pill degrades fully, ensuring that the created fractures remain open to flow.

Pre/Post Flush and Other Fluids

A small spearhead of HCl was used in Stage 1 only – 100 bbl of 20% HCl ahead of the first pad – to ensure breakdown of the perforations which had high perforation friction at low rates. Additionally, mutual solvent (~2,530 bbl across stages) was used as pre-flush and overflush to improve acid coverage and water-wet the formation (reducing emulsions). Finally, slick brine (2% KCl) was used for displacing fluids and for the initial injection tests; about 1,764 bbl of slickbrine was pumped during the main treatments for displacement and miscellaneous steps.

Cycle Design

The main acid fracturing in each stage was executed in multiple cycles of pad + acid + diverting agent. Based on the data from Stage 1 and design optimizations, the number of cycles and diverter usage were slightly different in each stage:

Stage 1

Designed for 8 cycles. Each cycle consisted of ~240 bbl pad followed by ~230 bbl SPRA, then ~110 bbl of VDA diverter acid. The stage began with a 100 bbl HCl spearhead (as noted) and ended with a final 15% HCl flush (70 bbl) to push the last of the acid out of the tubing. After four cycles, a single biodegradable diverter pill (DLPD) (~0.7 bbl of concentrated slurry) was pumped and flushed into the formation with brine. This pill was intended to seal fractures after treating roughly half the interval. The remaining four cycles were then pumped. A post-flush of mutual solvent and brine completed the stage. Stage 1 was initiated at 6 bpm (during spearhead and early pad) and gradually increased to ~20 bpm by the end of treatment.

Stage 2

Designed for 11 cycles (a longer interval with more perf clusters). No spearhead acid was used in Stage 2 – pumping began directly with pad at 10 bpm, as the formation was already fractured from Stage 1 and demonstrated injectivity. Each cycle in Stage 2 was of similar composition: roughly 240 bbl pad, 215 bbl of SPRA, 100 bbl VDA (slightly larger volumes than Stage 1. Two DLPD pills (each ~1.5 bbl slurry) were scheduled: the first after 3 cycles and the second after 4 more cycles (i.e. after cycle 7). The pills were displaced with brine at ~15 bpm and observed for pressure response. After the second pill, the remaining cycles were pumped to reach the total of 11 cycles. Rate: Stage 2's pumping rate was stepped up more aggressively – starting at 10 bpm and ramping to ~25 bpm by the final cycles. This higher rate regime was enabled by confidence in the well's pressure containment from Stage 1 and by the lower friction achieved with the pad fluid and single-phase acid (friction ~2,441 psi at 25 bpm, measured during flush).

Stage 3

Initially designed for 11 cycles similar to Stage 2. However, operational constraints (surface pressure limit) led to an earlier termination after 6 cycles in Stage 3 (details in Results). No spearhead was needed; pumping commenced directly with pad. The cycle volumes were somewhat reduced in this top stage: ~145 bbl pad, 135–140 bbl SPRA, and ~50–60 bbl VDA per cycle, reflecting the shorter perforated interval. One DLPD pill (~1.5 bbl) was planned after the first 4 cycles. Rate: Owing to the lower fracturing pressure expected (Stage 3 being shallower and already partially treated above), Stage 3 was started at a relatively high rate of 25 bpm. The pump rate was further increased to ~35 bpm during pad injection to extend the fracture quickly. This approached the surface equipment pressure limit (MASP), so the rate was subsequently throttled back (around the time of acid injection and pill placement) to keep surface pressure within safe limits (≈13,000 psi).

Results and Discussion

Stage 1 – Execution and Performance

Treatment Execution

Stage 1 covered the bottom-most Middle Marrat perforations (four clusters from 18,620 ft to 18,785 ft MD). Following the diagnostics, the main acid fracturing treatment was pumped as shown in Fig. 7 below showing the pressure and rate profile for the entire treatment. Starting with spearhead acid, Pumping then continued into the pad and acid cycles. As the pad fluid entered, surface pressure climbed steadily but remained below the expected fracture extension pressure until about 6 bpm was achieved. Once the fracture opened, the pressure stabilized and began rising more gradually with the increasing pumping rate. Over the course of 8 cycles, the rate was ramped from 6 bpm up to 30 bpm and ending with ~20 bpm by the last cycle. The peak treating pressure in Stage 1 reached ~12,992 psi at surface, well within the 13,000 psi MASP for the wellhead equipment. The DLPD pill was pumped after cycle 4 as planned. This was a small pill (0.71 bbl slurry). It was displaced to perforations at 6 bpm. Unlike later stages, the Stage 1 pill did not show a dramatic pressure increase when it reached the formation. The treating pressure plot exhibited at most a subtle uptick (on the order of <50 psi) during and after pill placement – an ambiguous indication of diversion. This muted response suggested that either the pill volume might have been insufficient to fully bridge the multiple open fractures, or that the created fracture geometry in Stage 1 was such that diversion was less effective (possibly a dominant fracture taking the most fluid despite the pill). Based on this, it was decided to increase the diverter volume and concentration for subsequent stages to ensure a more pronounced diversion effect. After the diverter, pumping resumed with cycle 5. The post-diverter cycles continued to show gradually increasing pressure as the acid reacted and leaked off. Notably, even though diversion success was uncertain, fluid efficiency remained reasonable – the pressure did not rapidly fall, indicating the fracture stayed open and accepting fluid through all cycles. Once all 8 cycles were completed, the job was stopped (shut-in) temporarily before the clean fluid cycle was squeezed into the formation at low rate. At shut-in, the instantaneous shut-in pressure (ISIP) was recorded as 9,334 psi at surface. Accounting for hydrostatic head and friction, this corresponded to an estimated bottomhole ISIP of ~15,868 psi, equating to a fracture gradient ~1.07 psi/ft. This gradient is slightly above the 0.84 psi/ft closure obtained in the mini-frac, as expected for a pressurized fracture at the moment of shut-in. The net pressure (difference between BHP and closure pressure) at ISIP was on the order of 1,800 psi, reflecting the presence of a large acid-etched fracture that was still held open by internal pressure.

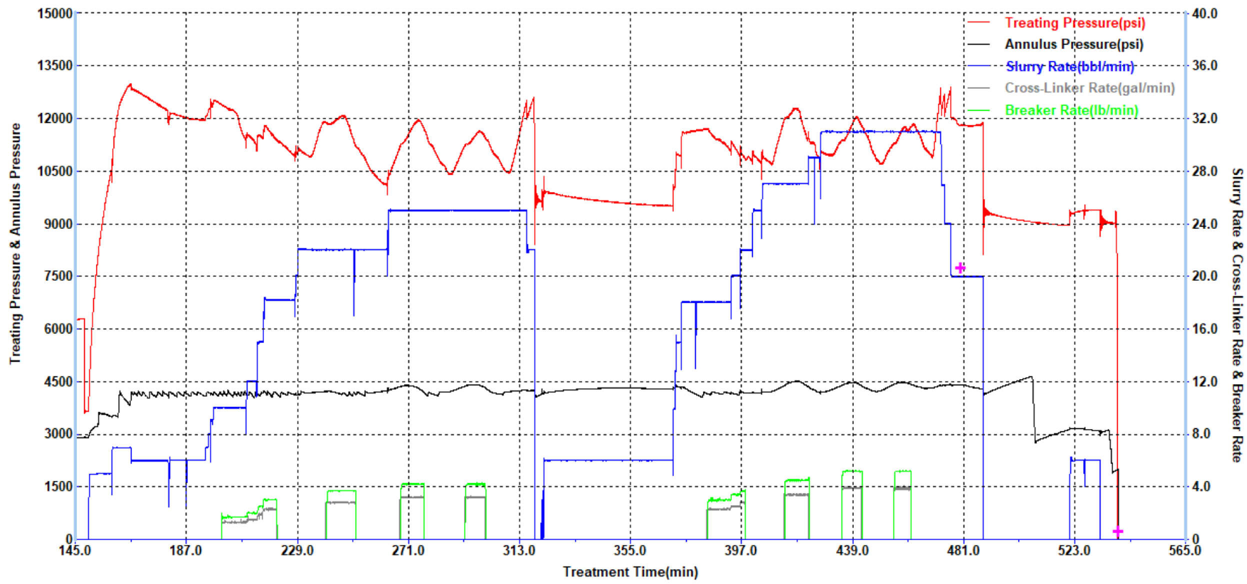

Figure 7—SA-XX Stage1 Acid Fracturing Treatment Plot

Operational learning: The Stage 1 results led to the recommendation to increase near-wellbore diverter volume and concentration in later stages to force improved distribution.

Stage 2 – Execution and Performance

Stage 2 covered a larger interval (five clusters from 17,935 ft to 18,350 ft MD), Stage 2's step-rate test with slickwater confirmed good injectivity – the well took up to 15 bpm with friction ~2,295 psi at surface and no anomalies. The ISIP after this brine injection was higher (11,905 psi surface). Treatment Execution: The Stage 2 main treatment was redesigned to incorporate two DLPD pills (versus one in Stage 1) and a higher initial rate, based on Stage 1 learnings. Pumping began directly with pad (no HCl spearhead), at 10 bpm. Pad and Acid Cycles: Stage 2 pumped a total of 11 cycles, as per Table 1. The rate was increased in steps: 10 bpm → 15 bpm during the first few cycles, then up to ~20 bpm by cycle 5, and ~25 bpm by cycle 11. Figure 8 illustrates Stage 2 treating pressure, rate versus time.

Table 1—Summary of wellbore access issues in SA-XX during plug & perf stimulation operations in Chronological Order

Depth (ft)	Run	BHA Maximum OD (inches)	Event
xz446	Tractor Conveyed Isolation Plug	3.445	Plug stuck
xy083	CT Conveyed Isolation Plug	3.445	Plug held up
xx553	CT Conveyed Milling Tool	3.5	BHA held up
xx565	CT Conveyed Milling Tool	3.4	BHA held up
xx565	CT Conveyed Milling Tool	3.5	BHA held up
xy119	CT Conveyed Milling Tool	3.4	BHA held up
xy120	CT Conveyed Lead Impression Block	3.4	BHA held up

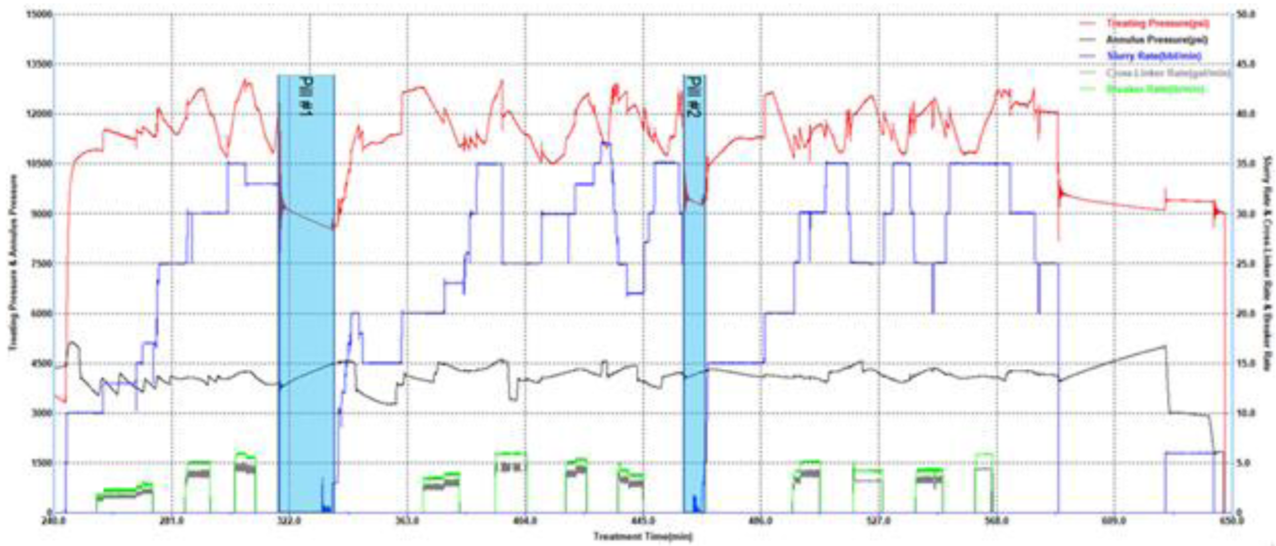

Figure 8—SA-XX Stage2 Acid Fracturing Treatment Plot

The pressure during pad injection of each cycle showed a repeating pattern: a low-pressure trend at the start of each new cycle (as fluid enters an open fracture), followed by a rise as the pad creates new fracture area, then a moderate drop when acid is introduced (acid etching reduces net pressure), and again a rise as acid spends and VDA enters (VDA gelling increases resistance). Diverter Effects: The first DLPD pill (approximately 1.5 bbl slurry) was pumped after the 3rd cycle. It was displaced at 15 bpm to ensure deep placement. Upon entering the formation, a notable pressure increase (~200 psi) was observed on the surface gauge. This positive "pill hit" indicated that the diverter effectively bridged and restricted the dominant flow paths, raising treating pressure until a new fracture segment took the fluid; when pad injection resumed for cycle 4, the treating pressure started from a higher baseline, confirming that the previous fractures were at least partially isolated. After cycle 7, the second DLPD pill (~1.5 bbl) was deployed. This pill yielded a smaller pressure response – about +71 psi at surface. The reduced bump could be due to a more complex fracture system, providing alternate leakoff paths. Nonetheless, there was a discernible pressure rise, suggesting some diversion. Based on these observations, the diverter strategy in Stage 2 was deemed effective. The recommendation was to maintain similar pill volumes and concentrations for subsequent stages, as the 200 psi jump from the first pill demonstrated strong diversion of acid into new zones.

Stage 3 – Execution and Performance

Stage 3 targeted the uppermost Middle Marrat clusters (three perforation sets from 17,540 ft to 17,815 ft MD). Stage 3's injectivity test showed a lower breakdown pressure – the formation parted at a surface pressure of only ~6,650 psi (ISIP), corresponding to ~0.85 psi/ft gradient. This is significantly less than the previous stages (by ~0.2 psi/ft), indicating either a lower stress state or perhaps some depletion in this upper interval. Closure gradient was ~0.81 psi/ft from the SRT. This meant Stage 3 could be fractured at lower pressure, but conversely, the lower stress could lead to wider fractures and potentially higher leakoff (due to potentially more connected porosity). Treatment Execution: Stage 3's treatment was begun with confidence to pump at a high rate from the start. The pad was immediately ramped to 25 bpm within the first minute of injection. Pressure response was as expected – relatively low initial pressure (~6,000 psi) until fracture initiation, then a sharp uptick as the fracture took fluid. Once the fracture was open, at 25–30 bpm the treating pressure rose into the 10,000–11,000 psi range. Pad volume of the first cycle was slightly lower than earlier stages, and the fracture extension was evident by a leveling-off of pressure. As acid was introduced, pressure dipped (due to acid etching reducing width), then climbed again as VDA was injected and began to divert. The rate was further increased to 35 bpm in the second cycle, marking the highest rate of the entire campaign. At 35 bpm, surface pressure approached ~12,900 psi – very near the MASP of 13,000 psi for the well. Consequently, the operations team decided to throttle the rate back to ~30 bpm to avoid risking pressure exceedance, especially as more cycles would continue to build pressure. This adjustment illustrates a key operational challenge: balancing the desire for high injection rates (to generate extensive fractures and overcome leakoff) with surface pressure limitations in deep wells.

Figure 9 shows the treating pressure plot for Stage 3, where the rate reduction can be seen after the early peak. Only 6 cycles were pumped in Stage 3, fewer than the planned 11. This early termination was primarily because the MASP was reached when pumping the 6th cycle at ~30 bpm – the treating pressure had climbed back to ~13,000 psi as the fracture network grew and fluid efficiency dropped somewhat. Rather than risk a screen-out or an uncontrolled pressure rise, the DLPD diverter pill was deployed after cycle 4 (one pill of ~1.5 bbl, as designed). At the time of pill displacement (done at 15 bpm to place it effectively while controlling pressure), a pressure increase of ~152 psi was observed – a clear indication that the pill bridged and diverted fluid. This was a larger response than seen for the second pill in Stage 2 and is likely because Stage 3's single pill had fully designed concentration in a smaller fracture set, creating a strong seal. The treating pressure baseline after the pill was noticeably higher, meaning the subsequent cycles (5 and 6) were likely stimulating previously under stimulated perforations or fracture wings. When pumping was halted at the end of cycle 6 (just before pad of cycle 7 was to start), the surface ISIP was recorded as 9,056 psi, which corresponds to ~15,591 psi bottomhole, or ~1.05 psi/ft. This is slightly lower than the ISIP gradients of Stages 1 and 2, which fits with the lower closure stress in Stage 3's interval.

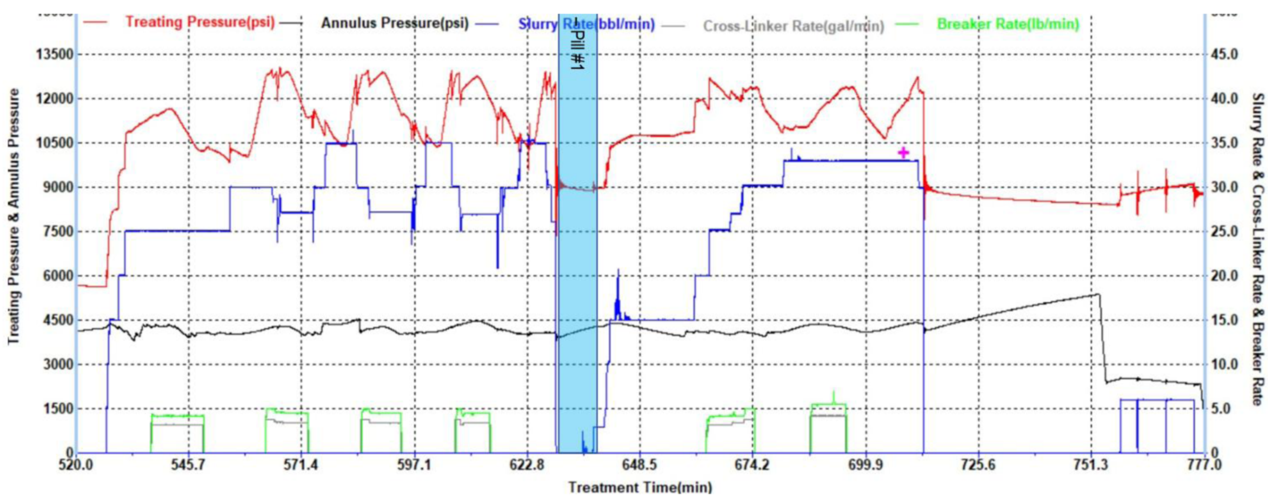

Figure 9—SA-XX Stage3 Acid Fracturing Treatment Plot

The net pressure at shut-in for Stage 3 was relatively high (since a lot of fluid was pumped and diversion had concentrated on the acid), helping ensure good fracture width at closure. Post-Treatment and Observations: Stage 3's limited number of cycles means slightly less acid was injected (1,500 bbl vs. 2,350 bbl in Stage 2), but it was still a significant treatment. The effectiveness of diversion in Stage 3 was perhaps the best of all: the 152 psi jump confirmed that the DLPD pill successfully segregated the treated zone, forcing new fractures to take the remaining fluid. This likely compensated for the fewer cycles by focusing the latter acid into fresh rock. The fluid efficiency in Stage 3 might have been somewhat lower (more leakoff) than Stage 2.

Post-Stimulation Outcomes

After completing the three stages, Well SA-XX had effectively been acid-fractured over a gross interval of ~1,200 ft (with ~300 ft of treated pay per stage on average). The well was flowed back and cleaned up. Although production data is beyond the scope of this paper, the immediate surface observations were positive: the well unloaded acids and diverter fluids easily (a benefit of using degradable, non-damaging diverters), and there were early indications of improved well productivity (surging flow of spent acid and reservoir fluids).

Key Learnings

Fluid System Synergy: The combination of a retarded acid + self-diverting acid + fiber/particle diverter proved highly effective for multi-zone stimulation. The single-phase retarded acid allowed high-rate pumping with reduced friction and deeper penetration (no emulsions or phase separation issues were encountered. The VDA ensured in-situ diversion and leakoff control, contributing to higher fluid efficiency and uniform acid distribution. The OPS-L pill provided a reliable near-wellbore seal when appropriately dosed, as evidenced by measurable pressure increases and subsequent improved acid coverage in later stages. These fluids complemented each other – the retarded acid did the primary etching, the VDA gelled in situ to divert within the fracture, and the

OPS-L pill diverted at the fracture entrance to initiate new fractures

Diversion Strategy Matters: Stage 1's minimal pill effect underscored that insufficient diverter volume can lead to suboptimal coverage. By enlarging the pill volume in Stage 2 (and adding a second pill), the team achieved clear diversion signals and likely treated more perforations. Thus, for thick carbonate reservoirs with many clusters, multiple diversion cycles are recommended to sequentially stimulate all zones. The use of degradable fibers to bridge fractures of various widths was crucial – pure particles might have passed through the large acid-created fissures, but fiber bridging created the necessary seal. The lesson is to err on the side of higher diverter concentration to ensure a robust pressure response, as it is easier to confirm diversion success through a pressure bump than to infer it from lack of one.

Rate and Pressure Management: Pumping at higher rates (20–35 bpm) was beneficial to extend fractures and offset leakoff, but it brought the operation close to equipment limits. A clear understanding of friction pressures and step-rate data allowed the team to push rates safely. In Stage 3, hitting MASP required halting early; planning a slightly lower max rate or having higher pressure- rated equipment could allow full design volumes in future treatments. In deep wells, having a margin between expected treating pressure and MASP is important – the job was successfully completed because this margin was respected and operations were adjusted in real time when pressures neared the limit.

Stage Isolation: While not the main focus of this paper, the importance of reliable stage isolation was highlighted by the sand plug attempt. Mechanical packers or drillable bridge plugs are preferred for isolating previously stimulated stages, as chemical/mechanical sand plugs may not hold in large-diameter, high-pressure conditions. In SA-0695, effective isolation between Stage 1, 2, 3 was achieved (likely via plugs/set packers by the workstring), but isolating Stage 3 for a potential Stage 4 failed with a sand plug. In future multi-stage acid fracs, a robust isolation plan (such as a retrievable bridge plug or inflatable packer) between stages can save time and ensure each stage is treated independently without cross-flow.

Data-Driven Design Adjustments: The iterative design approach – using Stage 1 results to refine Stage 2 (increasing diverter, modifying initial rate) and Stage 2 to refine Stage 3 (maintaining diverter strategy, anticipating lower pressure) – was critical for success. Early diagnostic data (closure pressure, fluid efficiency) fed directly into treatment volume calculations and prevented under/over-treatment. This case validates the DataFRAC approach: investing a small volume of fluid in testing yields data that can substantially improve the main treatment outcomes.

Geomechanics and Stress analysis (including 3D Far Field Sonic logging). The well SA-XX was drilled to TD utilizing Ultra Deep Azimuthal Resistivity (UDAR) geo-steering and near bit azimuthal Gamma Ray imaging. Slim High Definition Elemental Spectroscopy (HDES), Slim Cement Bond Map (SCBM) and Slim Dipole Sonic (SDST) tools were deployed once the lateral was cased with a 4.5 liner. HDES data provided valuable mineralogy data to perform petrophysical evaluation while the SCBM provided azimuthal cement bond quality in the liner. The purpose of acquiring SDST data was twofold:

- Extract representative Compressional & Shear Velocities from the wideband frequency data thereby providing invaluable velocity data for Geophysics & Geomechanics.
- Characterize the acoustic reflector surface geometry from extended listening time waveforms for Structural Geology applications with the 3DDFM workflows (Donald et. Al 2020)

Integration of UDAR, 3DDFM and HDES data shows that the lateral is placed in between two conductive-porous geo-bodies with variable Dolomitization, separated by a resistive-tight Calcite layers. The multi-physics measurements identified variations in fluid saturation in the porous geo-bodies and 17 faults that the lateral had intersected. Due to the structural complexity, at many places, the well trajectory is seen to be in the tight layer. As seen in Figure 10, 3DDFM data revealed sub layers within the geo-bodies and intervals where the stacked geo-bodies were connected thorough fractures (Al-Ajmi et. Al. 2024).

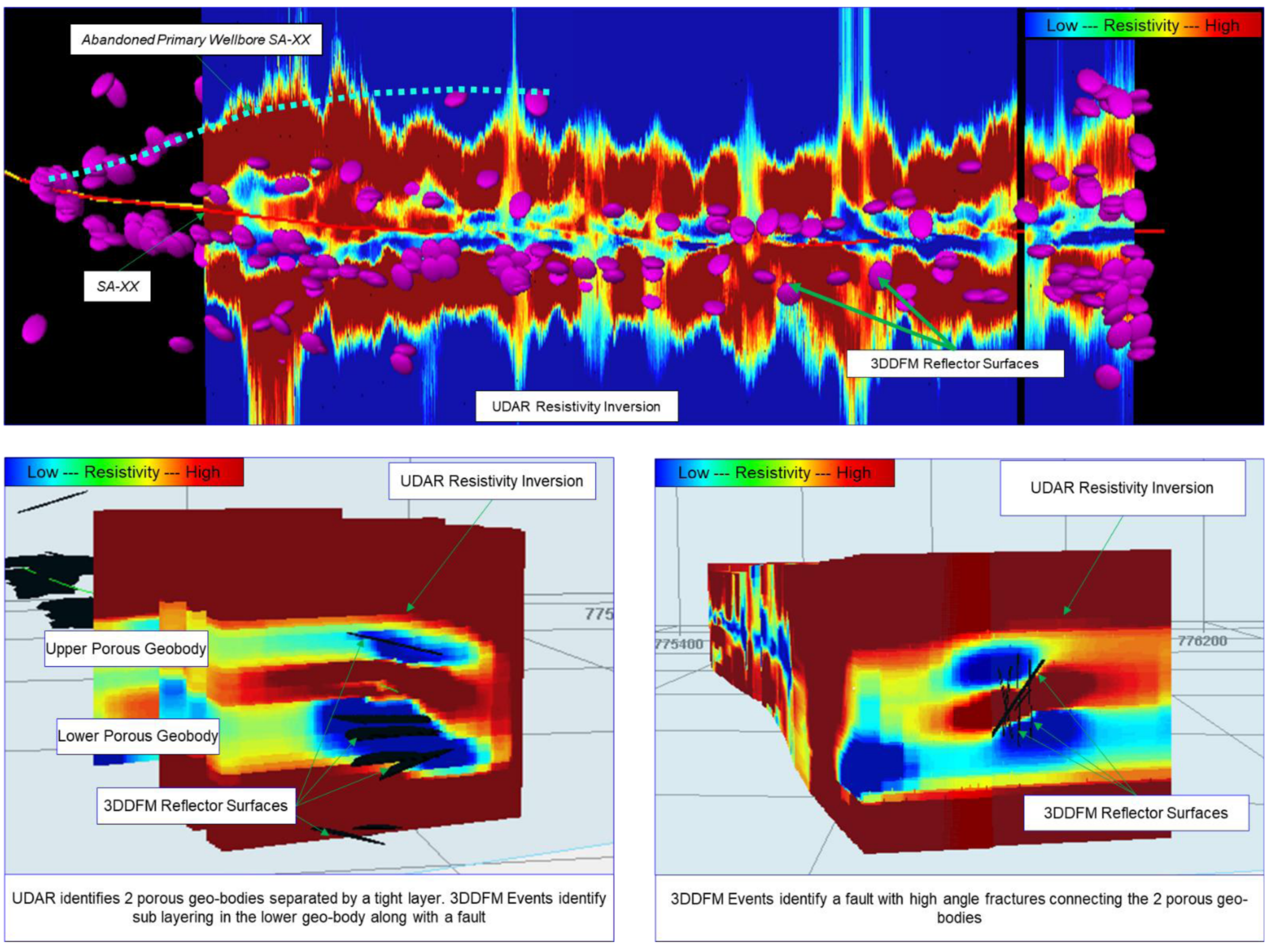

Figure 10—SA-XX UDAR & 3DDFM Data Integration Reveals Geological Complexity in FZ-7

Mechanical Earth Model (MEM) is a numerical representation of the state of in-situ stresses, pore pressure and rock mechanical properties for a specific stratigraphic section in a field or basin. It includes rock elastic and strength properties, pore pressure and in-situ earth stresses (Plumb. et al. 2000). Slowness data measured during the SDST survey was used to build an anisotropic MEM based on dipole shear wave anisotropy. The in-situ stress profile was used to define the Completion Quality (CQ) along the lateral length and was used to plan well stimulation operations. The CQ based approach was used instead of a geometric staging to determine the acid fracturing stages and perforation clusters due to the following reasons:

- Strongly anisotropic Strike-Slip regime – Many intervals exhibit 7000 psi stress anisotropy between Maximum and Minimum horizontal stresses. The intervals with high stress anisotropy are anticipated to have fracture initiation pressure exceeding the well completion and well head elements during stimulation treatment.

- Presence of tight calcite layers where a strong increase in closure stresses is seen. These intervals were anticipated to be poorly stimulated with probable height restricted induced fracture geometry.
- Presence of many faults and natural fracture clusters where unwanted fluid leak-off would reduce stimulation fluid efficiency.

Based on the MEM, Critically Stressed Fracture (CSF) analysis was performed on the structural events identified by the 3DDFM. CSF analysis helps in understanding the stress state acting on faults and fractures to characterize their mechanical behavior due to dynamic changes in pore pressure during drilling, stimulation and production from the well (Rogers, 2002). Faults and fractures are pre-existing discontinuities that can re-activate when the effective stresses along the fault/fracture plane exceed the shear strength of the rock. The discontinuity is then referred to as unstable. If the shear strength on the discontinuity plane is not exceeded, the discontinuity is referred to as stable under the current stress state. Structural features experiencing high shear stress to normal stress ratios are expected to be mechanically unstable and prone to slippage / re-activation failures. It was seen that intervals with low effective stresses (favourable CQ) coincided with intervals where the lower and upper porous geo-bodies are connected by critically stressed compliant fractures.

The initial acid fracturing stimulation design consisted of a total of 7 stages with 3 to 4 perforation clusters. Each stage was to be isolated with Coiled Tubing (CT) millable bridge plugs (Figure 11).

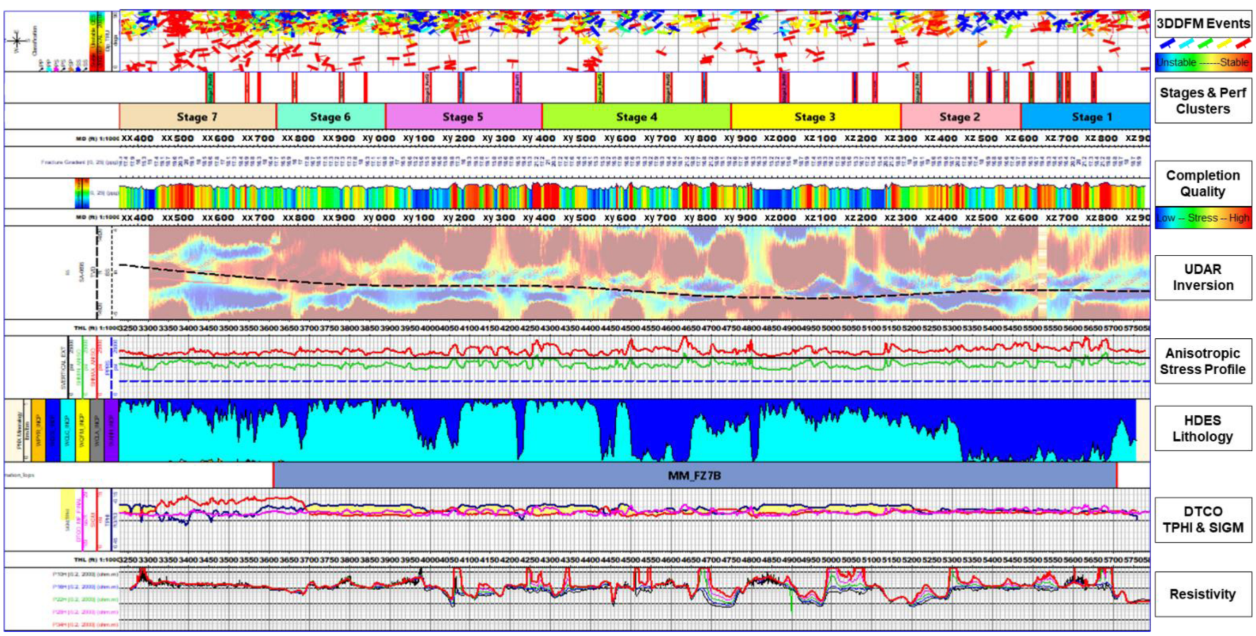

Figure 11—SA-XX Petrophysical Analysis & Mechanical Earth Model for Acid Fracturing Stimulation Design

During executing the 1st stage of the acid fracturing treatment, a good match between data-frac closure pressures and the MEM was seen indicating the robustness of the model. As anticipated, pressure dependent leak-off signatures were seen on the diagnostic plots. The main frac treatment for the 1st stage was subsequently re-engineered with biodegradable particulate diverter pills as mentioned previously.

Challenges in wellbore Access and Casing Deformation observation. Post Stage-1 acid fracturing treatment, the isolation plug for Stage-2 could not be conveyed to the desired depth and had to be set shallower. The stimulation treatment was re-designed to 6 stages. Over the course of operation for the other stages, multiple wellbore access issues were faced during plug setting and CT clean out operations (Table 1).

A total of 4 stages could only be executed eventually. To investigate the root cause of the wellbore access issues, a Multi Finger Caliper survey was carried out to investigate the possibility of casing deformation. The survey was done after executing Stage-4 stimulation treatment and the multiple failed CT interventions. Multiple casing deformations were observed correlating very well with the intervals with restricted wellbore access. The data was integrated with the MEM and the CSF analysis performed earlier (Figure 12).

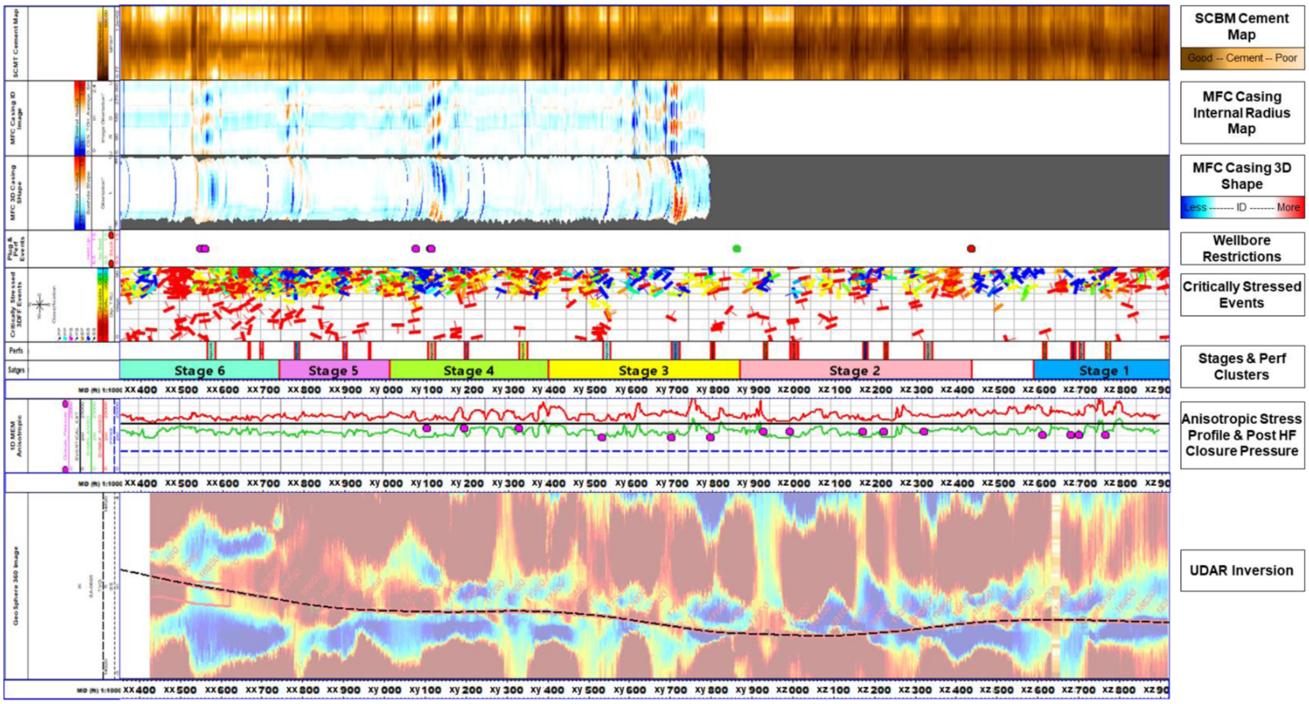

Figure 12—SA-XX Casing Deformation Analysis

The observations related to the casing deformations can be summarized below:

- Casing deformations and wellbore access restrictions, during tractor & CT conveyed runs post Stage-1, align very well with intervals with unstable critically stressed fractures.
- SCBM azimuthal cement map reveals that the cement is non-uniformly around the casing. The cement bond on the low side of the casing appears to be better than the high side of the casing. Majority of the deformations can be seen in intervals with poor azimuthal cement coverage.
- Casing deformations appear to initiate close to high stress anisotropy layers. They are not seen across perforated intervals.

Conclusion

A comprehensive three-stage acid fracturing campaign in a deep carbonate well (SA-0695, Middle Marrat Formation) was executed successfully, employing innovative fluid systems for enhanced diversion and penetration. Key conclusions include:

Successful Multi-Zone Stimulation: Stages 1, 2, 3 were pumped as per design and achieved acid fracturing of the intended intervals without any major operational issues. Over 1,500 ft of gross interval was stimulated in one continuous operation, which is an uncommon feat in such deep, high-pressure reservoirs. The maximum surface pressure observed was ~12,996 psi and the highest pump rate reached 37.2 bpm, demonstrating the ability to pump aggressive jobs in deep wells with proper fluids and equipment.

Fluid System Performance: The single-phase retarded acid (OPR-15) proved its value by extending live-acid contact and allowing high-rate fracturing. More than 5,700 bbl of this acid was effectively placed without phase separation issues. The acid's low friction enabled pump ramps to 30–35 bpm when needed. The viscoelastic diverting acid (VDA) provided in-situ diversion and leakoff control, with ~2,530 bbl pumped in total. Its contribution to fluid efficiency and distribution was evident in the smooth pressure trends and absence of rapid screen-out. The OPS-L biodegradable diverter was a critical component for near-wellbore diversion: a total of ~641.5 lb of diverter solids (fiber + particles) was used across the stages, resulting in clear pressure indications of successful diversion in Stages 2 and 3. The degradable nature of this system meant no remedial wellbore clean-up was necessary post-treatment.

Stage-wise Learnings: Stage 1 taught the importance of diverter volume; Stage 2 demonstrated effective multi-cycle diversion with two pills, leading to better coverage; Stage 3 showed the limits imposed by surface pressure and the need for real-time adjustments. Notably, the fracture closure gradients differed by stage (approx. 0.84, 0.94, 0.81 psi/ft for Stages 1, 2, 3 respectively), which likely reflects stratigraphic or depletion differences. All stages, however, achieved fracture gradients around 1.05–1.08 psi/ft at ISIP, meaning the designed pumping schedules were adequate to overcome the in-situ stresses in each case. Total pad volumes (YF125 pad ~6,180 bbl) and acid volumes were properly tailored to each stage's leakoff – evidenced by the lack of any premature screen-out and the fact that fractures could be propagated until acid was fully spent or pressure-limit reached.

Operational Outcomes: The well was successfully stimulated in three stages with minimal formation damage – no polymer gels or solid proppants were left in the formation. The high-rate, high-pressure pumping was executed without incident, showcasing good operational control. Minor challenges like sand plug failure highlight areas for improvement in stage isolation techniques. The overall treatment results are expected to significantly enhance well productivity, as the etched fractures extend deep into previously tight rock. Early post-stimulation data (not detailed here) reportedly showed substantial increases in well flow capacity, validating the stimulation approach.

Recommended Best Practices: For similar future jobs, we recommend: (1) Use of single-phase retarded acids for deep carbonate fracturing to achieve higher rates and deeper penetration

; (2) Inclusion of self-diverting acids (VDA) to improve zonal coverage and fluid efficiency

; (3) Deployment of fiber-assisted diverter pills between acid cycles for multi-cluster intervals

, with sufficient volume to ensure a clear pressure response; (4) Performing step-rate tests and mini-fracs on each stage to tailor pad and rate schedules to actual formation response; and (5) Ensuring robust stage isolation either via mechanical means or proven chemical plugs if a well has to be segmentally stimulated. Following these practices can markedly improve stimulation outcomes in deep, high-pressure carbonate reservoirs.

In conclusion, the Well SA-0695 case study illustrates how an engineered combination of new acidizing chemistries and diversion techniques can overcome the challenges of deep carbonate stimulation. The result was a technically successful multi-stage acid fracture treatment, with each stage building on the learnings of the previous one, ultimately leading to enhanced reservoir contact and expected improved hydrocarbon recovery from the Middle Marrat formation.

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